

Homework 8

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Problem 1. Let $\mathfrak{g}$ be a 5-dimensional Lie algebra over a field of characteristic 0. Suppose that with respect to a basis $\{e_1, \dots, e_5\}$, every ad_x is strictly upper triangular, and the pattern of nonzero entries is given by

$$\begin{pmatrix} 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 0 & * \\ 0 & 0 & 0 & 0 & * \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

where each “*” indicates an entry that is nonzero (for suitable $x \in \mathfrak{g}$), and all other entries are zero.

The **upper central series** is defined inductively by

$$Z_0 = 0, \quad Z_{i+1} = \{x \in \mathfrak{g} \mid [x, \mathfrak{g}] \subseteq Z_i\}.$$

The **lower central series** is defined by

$$\mathfrak{g}_1 = \mathfrak{g}, \quad \mathfrak{g}_{i+1} = [\mathfrak{g}, \mathfrak{g}_i].$$

1. Express each Z_i as the span of a subset of the basis vectors $\{e_1, \dots, e_5\}$.
2. Express each $\mathfrak{g}_i$ as the span of a subset of the basis vectors $\{e_1, \dots, e_5\}$.
3. Give an example of a nilpotent Lie algebra (not necessarily of dimension 5) for which the sequence $\dim Z_i$ is **not** the reverse of the sequence $\dim \mathfrak{g}_i$.

Proof. (1,2) We adopt the following notation: for each $x \in \mathfrak{g} \setminus \{e_5\}$, write

$$\text{ad}_x = \begin{pmatrix} 0 & 0 & x_{13} & x_{14} & x_{15} \\ 0 & 0 & 0 & x_{24} & x_{25} \\ 0 & 0 & 0 & 0 & x_{35} \\ 0 & 0 & 0 & 0 & x_{45} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad \text{ad}_{e_5} = \begin{pmatrix} 0 & 0 & a_{13} & a_{14} & a_{15} \\ 0 & 0 & 0 & a_{24} & a_{25} \\ 0 & 0 & 0 & 0 & a_{35} \\ 0 & 0 & 0 & 0 & a_{45} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The assumption that each “*” is nonzero for suitable $x \in \mathfrak{g}$ in particular implies that for each *, there is at least one $x \in \mathfrak{g}$ for which that component of ad_x is nonzer. In particular, there exists $x \in \mathfrak{g}$ such that $x_{35} \neq 0$ or $x_{45} \neq 0$.

Let $x \in \mathfrak{g}$. Since $\{e_1, \dots, e_5\}$ is a basis, write $x = \sum_i c_i e_i$ for coefficients $c_i \in \mathbb{K}$, with $\mathbb{K}$ the base field. Then we have

$$[e_5, x] = \sum_{1 \leq i \leq 5} c_i [e_5, e_i].$$

Note that for any $y \in \mathfrak{g}$, the structure of the matrix ad_y implies $\text{ad}_y(e_1) = \text{ad}_y(e_2) = 0$. Also note that $[e_5, e_5] = 0$. Thus taking $y = e_5$ simplifies the above sum to

$$[e_5, x] = c_3 [e_5, e_3] + c_4 [e_5, e_4] = c_3 \text{ad}_{e_5}(e_3) + c_4 \text{ad}_{e_5}(e_4).$$

Note that $\text{ad}_{e_5}(e_3) = a_{13}e_1$ and $\text{ad}_{e_5}(e_4) = a_{14}e_1 + a_{24}e_2$. In particular, $\text{ad}_{e_5}(e_3) + \text{ad}_{e_5}(e_4)$ is linearly independent from $\{e_3, e_4\}$, and thus contains no e_3 - or e_4 -dependence. Thus $[e_5, x] = -[x, e_5]$ contains no e_3 - or e_4 -dependence. Therefore

$$-[x, e_5] = -\text{ad}_x(e_5) = -(x_{15}e_1 + x_{25}e_2 + x_{35}e_3 + x_{45}e_4)$$

must not depend on e_3 or e_4 , implying $-x_{35} = -x_{45} = 0$. Thus $x_{35} = x_{45} = 0$. Since $x \in \mathfrak{g}$ was arbitrary, for all $x \in \mathfrak{g}$, we have $x_{35} = x_{45} = 0$, contradicting the assumptions of the problem. Either the problem assumes a false statement is true, implying that all statements are true (including the claims of the problem), or $\mathfrak{g}$ does not exist, implying (1) and (2) vacuously. We proceed with the second assumption, and prove (3).

(3) Let $\mathfrak{g}$ a 1-dimensional Lie algebra and $\mathfrak{h}$ the Heisenberg algebra generated by the elements

$$X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

We show that $\mathfrak{k} := \mathfrak{g} \oplus \mathfrak{h}$ is an example of (3). Since $\mathfrak{h}$ has dimension 3, $\mathfrak{k}$ has dimension 4. Let $e \in \mathfrak{g}$ be the basis, such that $\mathfrak{k}$ has the basis

$$\{e, X, Y, Z\}.$$

Here we note that $[e, \mathfrak{h}] = 0$ (since we are identifying e with the ordered pair $(e, 0)$, and similarly for the elements of $\mathfrak{h}$). We also recall the commutation relations of $\mathfrak{h}$:

$$[X, Y] = Z, \quad [X, Z] = 0, \quad [Y, Z] = 0.$$

Since the only nonzero commutator between basis elements is $[X, Y] = Z$, we immediately see that $\mathfrak{g}_2 = [\mathfrak{g}, \mathfrak{g}] = \text{span}\{Z\}$ has dimension 1. Since the lower central series is indexed starting at 1, and the upper central series is indexed starting at 2, it suffices to show $\dim Z_1 \neq 1$.

Recall $Z_1 = \{x \in \mathfrak{k} \mid \text{ad}_x = 0\}$. We immediately see that $\mathfrak{g}$ must be abelian since $[e, e] = 0$, so we immediately have $\text{ad}_e = 0$. Additionally, we have $\text{ad}_Z(X) = \text{ad}_Z(Y) = 0$ and $\text{ad}_Z(e) = 0$, implying $\text{ad}_Z = 0$. Thus we have at least two linearly independent elements $e, Z \in Z_1$, implying $\dim Z_1 \geq 2 \neq 1$. Therefore $\mathfrak{k}$ is an example of such an algebra. ■

Proof. I also solve (1,2) in the intended way. For all i , note that $\mathfrak{g}_i$ and Z_i are all clearly subspaces of $\mathfrak{g}$, and note that we trivially have $Z_i \subseteq Z_{i+1}$ and $\mathfrak{g}_i \supseteq \mathfrak{g}_{i+1}$.

(1) Let $x \in \mathfrak{g}$; with notation as above,

$$\text{ad}_x(e_1) = 0, \quad \text{ad}_x(e_2) = 0,$$

$$\text{ad}_x(e_3) = x_{13}e_1, \quad \text{ad}_x(e_4) = x_{14}e_1 + x_{24}e_2,$$

$$\text{ad}_x(e_5) = x_{15}e_1 + x_{25}e_2 + x_{35}e_3 + x_{45}e_4.$$

We assume that for all “*” in the given form, there exists x such that that component is nonzero. Note that $Z_1 = \{x \in \mathfrak{g} \mid \text{ad}_x = 0\}$. Linear independence of e_1, e_2, e_3, e_4 and the above equations imply $Z_1 = \text{span}\{e_1, e_2\}$. Then Z_2 is the set of all x for which $\text{ad}_x(y) \in \text{span}\{e_1, e_2\}$ for all $y \in \mathfrak{g}$. The above equations imply this is $Z_2 = \{e_1, e_2, e_3, e_4\}$. By the same logic, we see $Z_3 = \text{span}\{e_1, \dots, e_5\}$.

(2) Note that $\mathfrak{g}_2 = [\mathfrak{g}, \mathfrak{g}]$. Letting $x \in \mathfrak{g}$ such that $x_{13} \neq 0$, and using that $\mathfrak{g}_2$ is a subspace, we first see that $x_{13}^{-1}\text{ad}_x(e_3) = e_1 \in \mathfrak{g}_2$. Now let x have $x_{24} \neq 0$. We have

$$x_{24}^{-1}(\text{ad}_x(e_4) - x_{14}e_1) = e_2 \in \mathfrak{g}_2,$$

since $e_1 \in \mathfrak{g}_2$ and thus $\mathfrak{g}_2$ is closed under subtraction by it. From this we similarly obtain $x_{35}e_3 + x_{45}e_4 \in \mathfrak{g}_2$. We cannot obtain $e_5 \in \mathfrak{g}_2$, so $\mathfrak{g}_2$ has dimension at most 4. I cannot find how to conclude that $e_3, e_4 \in \mathfrak{g}_2$, but the statement of (3) implies that this must be true. If one assumes that x_{35} and x_{45} are independent parameters this follows, but the problem does not state that. We proceed with the assumption that $\mathfrak{g}_2 = \text{span}\{e_1, \dots, e_4\}$, but note that if one takes $\mathfrak{g}_2 = \text{span}\{e_1, e_2, e_3 + ce_4\}$ for some $c \in \mathbb{K}$, then the logic below does not change by bilinearity of the bracket.

Now we immediately see that $\mathfrak{g}_3 = [x, y]$ with $x \in \mathfrak{g}$ and $y \in \text{span}\{e_1, \dots, e_4\}$. Recall that $\text{ad}_{e_1} = \text{ad}_{e_2} = 0$, so we get no information there. We then have that there is x such that $[x, e_3] = x_{13}e_1$, so $e_1 \in \mathfrak{g}_3$. We then obtain that $e_2 \in \mathfrak{g}_3$ by $e_2 = x_{24}^{-1}[x, e_4] - x_{24}^{-1}x_{14}e_1$ for suitable x . Additionally,

$$[e_5, e_3] = \text{ad}_{e_5}(e_3) \in \text{span}\{e_1, e_2\} \ni \text{ad}_{e_5}(e_4) = [e_5, e_4].$$

Thus $\mathfrak{g}_3 = \text{span}\{e_1, e_2\}$, since we have exhausted all combinations of basis elements. Finally, since $\text{ad}_{e_1} = \text{ad}_{e_2} = 0$, we obtain $\mathfrak{g}_4 = 0$. ■